Exercise 2 - Aggregate Functions on Grouping

```
SELECT *
FROM employees;
```

id	first_name	last_name	department	salary	hire_date	city
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San Francisco
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-07	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San Fran

Write a SQL query to find the total number of employes in the company

```
SELECT Count(*) AS total_employees
FROM employees;
```

total employees
10

Write a SQL query to find the total salary paid to all employees in the IT departments

```
SELECT SUM(salary) AS total_IT_salary
FROM employees
WHERE department = 'IT'
```

total_IT_salary
220 000

Write a SQL query to calculate average salary of employees in the HR department

```
SELECT AVG(salary) AS avg_HR_salary
FROM employees
WHERE department = 'HR'
GROUP BY department;
```

avg_HR_salary
495000

~~department~~
~~HR~~

Write a SQL query to find the highest and lowest salary in the company

```
SELECT max(salary) AS max_salary
       MIN(salary) AS min_salary
FROM employees;
```

max_salary	min_salary
62000	48000

Write a SQL query to count how many employees work in each city

```
SELECT city,
        COUNT(id) AS num_of_employees
FROM employees
GROUP BY city;
```

city	num_of_employees
New York	1
Chicago	3
LosAngeles	2
San Francisco	2
Houston	1

Write a SQL query to group employees by department calculate average salary in each department, and order the results in descending order of average salary

```
SELECT department,
        AVG(salary) AS avg_salary
FROM employees
GROUP BY department,
ORDER BY avg_salary DESC;
```

department	avg_salary
Finance	59 000
IT	55 000
Marketing	52 000
HR	49 000

Write a SQL query to list cities where more than one employee works, order by the number of employees in descending order

```
SELECT city
       COUNT(id) AS num-employees
FROM employees
GROUP BY city
ORDER BY num-employees;
```

City	num-employees
Chicago	3
San Francisco	2
LosAngeles	2
Houston	1
New York	1

Write a SQL query to find the department with the highest average salary

```
SELECT department,
       AVG (salary) AS avg-salary
FROM employees
GROUP BY department
ORDER BY avg-salary
LIMIT 1;
```

department	avg-salary
Finance	59 000

Write a SQL query to find department where the total salary exceeds 100 000

```
SELECT department,
       SUM (salary) AS total-salary
FROM employees
WHERE SUM (salary) > 100 000
GROUP BY department;
```

department	total-salary
IT	220 000
Finance	119 000
Marketing	105 000